

Goal. Use integration to find the volume of various *solids of revolution*.

Overview. Let $f(x)$ be a positive, continuous function. If the area underneath $f(x)$ from $x = a$ to $x = b$ is rotated around the x -axis, a 3-dimensional object is formed, and the volume of that object is given by the formula,

$$V = \pi \int_a^b f(x)^2 dx$$

1. Explain why the area of a cross section at $x = a$ is given by $\pi f(a)^2$. Then explain why the whole formula is valid.

2. Consider rotating the region under the function $f(x) = x$ from $x = 0$ to $x = 4$ about the x -axis.
 - (a) Sketch a picture of the solid. What type of solid is it?

 - (b) Use an integral to compute the volume of the solid.

 - (c) Use a known volume formula (you can Google one) to verify your answer from (b) is correct.

3. Consider the solid obtained by revolving the region enclosed by $y = 32 - 2x$, $y = 2x + 4$, and $x = 0$ about the x -axis.

(a) Sketch the solid.

(b) What are the vertical cross-sections of this solid?

(c) How can we modify the volume formula at the top of the worksheet to find the volume of this solid?